

Stefan Seman

Head of AI · ML Engineer · PhD Researcher

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PROFILE

AI leader and hands-on engineer. I run an AI and Data practice as **Head of AI** and still build the systems myself — LLM and RAG architectures, agentic applications, enterprise document intelligence, and cloud-native data platforms taken from problem statement to something running, monitored, and maintained. In parallel I deliver AI work through my own B2B consultancy for partner organisations in Europe and the US. Ten years of applied ML, grounded in a PhD track and **14 peer-reviewed publications** in computational cardiology and exercise physiology.

TECHNICAL SKILLS

LANGUAGES	Python, SQL, Bash, LabVIEW	ML & GENAI	PyTorch, Transformers, LangChain, LlamaIndex, RAG, AI Agents, Fine-Tuning, Embeddings, XGBoost, Computer Vision
BACKEND	FastAPI, Flask, REST APIs, Docker, PySpark, RabbitMQ	AZURE	AI Foundry, Azure OpenAI, AI Search, Data Factory, Synapse, Microsoft Fabric, Functions, Data Lake, Databricks, DevOps
AWS	Bedrock, Lambda, Step Functions, S3, ECS, EC2, SAM	DATA & VECTORS	PostgreSQL, DynamoDB, Redis, Qdrant, Pinecone, Milvus, PgVector
OPS & TOOLING	Git, Jenkins, Grafana, KubeFlow, Power BI	RESEARCH	Biostatistics, SPSS, ECG & CPET analysis, in silico trials

EXPERIENCE

Head of AI Apr 2025 – Present
ExpertGroup · Belgrade, Serbia

- Lead the AI and Data team, reporting to the CEO — organisation, people management, hiring, and mentoring.
 - Own AI initiatives end to end: originating projects, presales, architecture, and engineering standards.
 - Delivered RAG chatbots over live document collections — multilingual retrieval with answer lineage and real-time index updates.
 - Built gradient-boosted time series models for supply chain and sales forecasting on a monthly planning cycle.
 - Architecting crowd movement prediction and venue analytics for large-scale events.
 - Stay hands-on — demo pitches, proofs of concept, and MVPs built alongside the team.
- Stack:** Python, FastAPI, LangChain, Qdrant, RabbitMQ, Azure AI Foundry, Microsoft Fabric, Azure DevOps, Docker, XGBoost, Computer Vision

Independent AI & ML Consultant Oct 2024 – Present
404 Solutions · own B2B consultancy

- Deliver AI and software work for delivery partners in Europe and the US, serving their enterprise clients.
 - Designed and operated multi-tenant AI systems — document intelligence for automated extraction, classification, and summarization.
 - Built agentic graph RAG architectures for multi-document summarization and risk analysis in specialty insurance.
 - Built scalable AWS workflows on S3, Step Functions, Bedrock, and APIs supporting multiple organisations.
 - Deployed, monitored, and maintained production systems across distributed client environments.
- Stack:** Python, FastAPI, RAG, AI Agents, LangChain, AWS Bedrock, Lambda, Step Functions, S3, ECS, DynamoDB

Senior Data Scientist & Software Engineer in AI, R&D Oct 2023 – Sep 2024
Publicis Sapient

- Led development from proof of concept through to production-grade systems.
 - Built enterprise RAG systems with retrieval optimization, evaluated for accuracy and consistency rather than fluency.
 - Delivered an enterprise AI code copilot grounded in internal code and documentation, with retrieval respecting existing access boundaries.
 - Delivered client-facing solutions and technical propositions.
- Stack:** Python, FastAPI, LangChain, Transformers, PyTorch, Azure OpenAI, KubeFlow, Redis, Milvus, Docker

Data Scientist & Developer → Senior Specialist

May 2022 — Oct 2023

Valcon · Netherlands

- Led and mentored a team of six engineers.
- Owned the design of document processing microservices.
- Built data extraction and document classification services from scratch, delivered to production.

Stack: Python, Azure DevOps, Azure Data Factory, Synapse, Data Lake, PyTorch, Flask, Docker

Data Scientist

Oct 2021 — Apr 2022

Optimal Systems GmbH

- Model evaluation and hyperparameter tuning for ML classification systems.
- Built an Azure extraction and transformation platform feeding self-service Power BI reporting.

Stack: Python, Azure Data Factory, Synapse, Data Lake, Power BI, Docker

Researcher

Mar 2020 — Mar 2022

BioIRC — Bioengineering R&D Center

- Built components for SILICOFM, an EU Horizon 2020 computational platform for in silico clinical trials.
- ECG, CPET, and biomedical data processing; contributed to peer-reviewed publications.

Stack: Python, PyTorch, Statistical Analysis, SPSS, ECG, CPET

PhD Researcher

Oct 2018 — Mar 2022

University of Belgrade

- Developed research methodology and statistical frameworks; applied machine learning to biomedical datasets.
- Published internationally in cardiology and exercise physiology journals.

Stack: Python, PyTorch, SPSS, Machine Learning, Biostatistics

Sports Scientist

Oct 2014 — Jul 2018

Sport and Recreation Center Obliquis

- Built algorithms for signal processing and data collection; functional testing and performance analysis for athletes.

Stack: Python, SPSS, LabVIEW, Data Analysis

SELECTED PROJECTS

Document Chatbot with Live Lineage

ExpertGroup · team lead & delivery

High-precision multilingual chatbot over a living document set — every answer traceable to its source passage, index updated in real time as documents change.

Python · FastAPI · LangChain · Qdrant · Azure AI Foundry

AI-Powered Document Summarizer

404 Solutions · with Provectus

Summarization and point-of-interest extraction across large multi-document sets, on an agentic graph RAG architecture that holds context where chunking breaks it.

Python · AWS Bedrock · Step Functions · LangChain · LlamaIndex

Crowd Analytics & Movement Prediction

ExpertGroup · architect & technical lead

Crowd movement prediction for a large-scale public event — forecasting where density is about to build so visitors can be routed around congestion instead of into it.

Python · Computer Vision · Time Series · Azure IoT Hub · Fabric

Large-Scale Document ETL Pipelines

404 Solutions · with Provectus

Serverless ingestion and transformation moving documents at scale through load, parse, transform, and delivery — the substrate the summarization and risk systems run on.

Python · AWS Lambda · Step Functions · DynamoDB · S3 · SAM

Agentic AI Insights Platform

404 Solutions · with WiserTech

Orchestration graphs coordinating multiple agents across summarization, suggestion, and meeting preparation — surfacing what matters before it is asked for.

Python · AI Agents · RAG · Speech-to-Text

RiskFocus — Risk Summariser

404 Solutions · with Provectus

Multi-document risk extraction surfacing where documents reinforce, qualify, or contradict each other — with traceable lineage back to every source.

Python · AWS Bedrock · Lambda · ECS · RAG

Slingshot — AI Code Copilot

Publicis Sapient

Enterprise AI copilot taken from proof of concept through MVP to delivered product, grounded in internal code and docs with access-aware retrieval.

Python · Azure OpenAI · LangChain · Milvus · KubeFlow

SILICOFM — EU Horizon 2020

BioIRC · silicofcm.eu

Computational platform for in silico clinical trials of familial cardiomyopathies — ECG and CPET analysis, statistical modelling, patient-specific 3D heart models.

Python · PyTorch · Biostatistics · ECG · CPET

14 PEER-REVIEWED WORKS	138 CITATIONS	5 H-INDEX	3 FIRST-AUTHOR WORKS
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Sacubitril/valsartan for the treatment of non-obstructive hypertrophic cardiomyopathy: an open-label randomized controlled trial (SILICOFCM)
[European Journal of Heart Failure](#) · 2024 · co-author · 26 citations

The ventilatory efficiency parameters outperform peak oxygen consumption in monitoring therapy effects in patients with hypertrophic cardiomyopathy
[Progress in Cardiovascular Diseases](#) · 2024 · first author

A Machine Learning Approach to Gene Expression in Hypertrophic Cardiomyopathy
[Pharmaceuticals \(MDPI\)](#) · 2024 · co-author

Defining the importance of stress reduction in managing cardiovascular disease — the role of exercise
[Progress in Cardiovascular Diseases](#) · 2022 · co-author · 72 citations

Physical activity and exercise as an essential medical strategy for the COVID-19 pandemic and beyond
[Experimental Biology and Medicine](#) · 2021 · first author

Reduced cardiac workload and enhanced oxygen extraction during maximal exercise in recreational endurance runners
[Open Heart](#) · 2026 · co-author

EDUCATION

PhD Candidate, Sport Science
University of Belgrade · 2018 — Present
Machine learning applications in exercise physiology and biomedical signal processing.

Master's Degree, Sports Medicine
University of Novi Sad

Bachelor's Degree, Sport Science
University of Belgrade

LANGUAGES

- ▶ Serbian — native
- ▶ English — fluent

CERTIFICATIONS

- ▶ Microsoft Certified: Azure AI Associate
- ▶ Microsoft Certified: Azure Data Fundamentals
- ▶ Lakehouse Fundamentals — Databricks Academy
- ▶ Generative AI Fundamentals — Databricks Academy
- ▶ COVID-19 Contact Tracing — Johns Hopkins University

MEMBERSHIPS

- ▶ ACSM — American College of Sports Medicine
- ▶ SEBM — Society for Experimental Biology and Medicine
- ▶ Serbian Society for Prevention of Chronic Noncommunicable Diseases
- ▶ Serbian Mountain Rescue Service